

ETHAN MOON

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Education

University of Michigan College of Engineering

Bachelor of Science in Engineering, Major in Computer Science, Minor in Math

Ann Arbor, MI

Expected May 2028

- **Relevant Coursework:** Programming, Discrete Mathematics, Proof-Based Linear Algebra, Multivariable Calculus

Technical Skills

Languages: TypeScript, JavaScript, Python, C++, SQL, Lua

Frameworks: React, React Native/Expo, Next.js, Node.js, Hono

Data & Infrastructure: PostgreSQL, Drizzle ORM, Prisma, OpenAPI/Orval, Docker, Git

Engineering & Research: OpenFOAM, ParaView, NASA GMAT, Arduino, ESP32

Spoken Languages: English & Korean (Native/Bilingual); Spanish (Limited Working)

Experience

Lead Full-Stack Engineer, Ody

May 2026 – Present

- Promoted to Lead Engineer and Head of U.S. Team, effective September 2026, after ranking first among six interns despite being the youngest; conducted **10+** interviews to build the fall intern cohort. Currently managing and leading **10+** university students from **Stanford University, Oxford University, Carnegie Mellon, UC Berkeley, etc**
- Led the end-to-end redesign of a consumer social app and delivered merchant workflows for AI-assisted menu translation, custom-domain email campaigns, automated discounts, and multi-location analytics. Now tasked to configure deliveries using Uber direct API and an in-dashboard resetaurant management AI agent
- Merged **187** PRs from May–August 2026 across six client applications and the backend, owning features end-to-end from database design and APIs through React/Expo interfaces.

Undergraduate Researcher, University of Michigan

August 2026 – Present

Intelligent Robotics & Autonomy Laboratory (iRaL)

Ann Arbor, MI

- Investigating foundation models for onboard vision-language navigation and human-robot coordination on morphable drones, using hardware-in-the-loop testing to evaluate autonomy before real-world flight.
- Working with Professor Vasileios Tzoumas to explore commercializing the team's aerial-robot autonomy research through a potential robotics startup.

Researcher & Co-Author, Grand Valley State University

September 2025 – September 2026

- Developed non-Newtonian, multiphase OpenFOAM models across 32 bioink-deposition cases and trained a success-boundary surrogate that reached 84.6% leave-one-out accuracy on 13 labeled runs.
- Engineered a Dockerized Python/JavaScript research platform that converts VTK outputs into interactive dashboards, automating analysis of 244,800 raw CFD records across 16 runs and 51 timesteps each.
- Co-authored a 12-page peer-reviewed ASEE paper on a closed-loop bioprinting system integrating CFD, machine vision, and sensor feedback; now co-authoring two bioprinting journal manuscripts.

Projects

NoSpoilers – Full-Stack Movie Discovery Platform

NoSpoilers.xyz – June 2026 – Present

- Co-built and launched a movie discovery platform used by **80+** users, featuring collaborative movie picking, personalized Movie DNA profiles, and a privacy-first spoiler-blocking Chrome extension.
- Engineered the platform across **120 HTTP handlers, 43 Prisma/PostgreSQL models, and 42 user-facing pages**, with Neon PostgreSQL, cross-platform watchlist syncing, and full-stack ownership across backend APIs and frontend experiences.

ViolinTwin – AI Practice Coach, React Native/Expo

June 2026

- Built an Expo app that records violin practice audio, segments notes, and compares pitch and rhythm against scales or imported MusicXML/MIDI targets; added a Swift/Kotlin bridge for low-latency tuner events.

AcademiQ – Constraint-Aware Academic Planning Platform

May 2026

- Built a degree-planning engine that generates explainable eight-semester schedules across fastest, lowest-stress, and most-flexible strategies while enforcing prerequisites, course availability, credit limits, and internship terms with deterministic fallbacks in a React/tRPC/Drizzle application, validated by 43 unit and integration tests.

Selected Publications

- J. C. Chikwe, **E. Moon**, and A. Balsamy Kamaraj, “Building a Closed-Loop Bioprinting System Through Multi-Level Mentorship: Integrating High School, Undergraduate, and Graduate Students in Interdisciplinary Research.” *2026 ASEE North Central Section Conference*, 2026. [Paper](#)
- **E. Moon**, “Asian Carp Are Ruining the Great Lakes.” *International STEM Research Journal*, Issue 4, December 2024. [Article](#)

Honors & Awards

DECA State Champion · Dartmouth Book Award · **2nd Place, Engineering**, GYM International STEM Innovation Challenge · Finalist, International Astronomy & Astrophysics Competition